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Lab 04 – Chihuahua or Muffin Workshop

In this lab, we worked through the “Chihuahua or Muffin” workshop, which focused on image classification using machine learning. The main objective of the workshop was to build a neural network that could look at an image and determine whether it was a chihuahua or a muffin. At first, that sounds simple, but the entire process showed how much work happens behind the scenes before a model can make even a basic prediction. We cloned a repository, loaded image data, defined transforms, built a neural network using PyTorch, trained the model, and evaluated its performance. The overall goal was to understand how images are represented numerically and how neural networks learn patterns from that data.

One of the biggest concepts reinforced in this lab was image classification. Instead of the computer “seeing” a dog or a muffin the way we do, it sees pixel values arranged in a structured format. Each image was resized to a fixed dimension and converted into tensors. That resizing step was important because neural networks require consistent input dimensions. We defined the input height and width, and from there calculated the number of features going into the first linear layer. That helped me better understand how the total number of input features is literally the number of pixels times the number of color channels.

Another key concept was how a neural network processes data through layers. In our model, we used multiple fully connected layers with ReLU activation functions. The forward pass flattened the image into a single vector, passed it through hidden layers, and then produced output probabilities using softmax. Seeing how the data flows through each layer made the abstract idea of neural networks more concrete. This lab also reinforced how backpropagation and optimization work together. The optimizer adjusts weights based on loss, and over multiple epochs, the model improves its predictions.

Transfer learning was mentioned as a broader concept even though we built a simple model from scratch in this workshop. It made me realize that in real-world scenarios, instead of building everything from the ground up, developers often start with pre-trained models and fine-tune them for specific tasks. That approach saves time and computational power. Understanding the difference between building a custom network and using transfer learning gave me a better appreciation for practical machine learning

workflows.

There were definitely challenges in this lab. One challenge was correctly defining the image dimensions and connecting them properly to the neural network. If the height and width were wrong, the number of input features would not match the expected size in the first linear layer. That forced me to think carefully about how dimensions flow through the model. Another challenge was working through placeholder values in the notebook. The interactive format required careful reading and attention to detail. Instead of just running everything automatically, I had to understand what each section was doing before moving forward. That actually helped reinforce the learning process.

One major insight I gained is how easy it is for a model to memorize data instead of truly learning patterns. The separation between training and validation datasets showed why evaluation on unseen data is critical. Without a validation set, a model could appear accurate while simply memorizing training examples. This reinforced the importance of proper dataset splitting and evaluation metrics in machine learning projects.

In terms of real-world applications, image classification has huge potential. It can be used in medical imaging to detect diseases, in security systems for facial recognition, in manufacturing for defect detection, and in self-driving cars for object recognition. Even something that seems simple like distinguishing a dog from a muffin demonstrates the underlying principles used in advanced AI systems. This lab helped me connect classroom concepts to practical uses in industries like healthcare, cybersecurity, and automation.

Personally, this lab helped me feel more confident working with deep learning frameworks like PyTorch. At first, neural networks can seem overwhelming because of the mathematical background behind them. But breaking it down step by step—loading data, defining transforms, building layers, training, and evaluating—made it manageable. It also showed me that debugging and problem-solving are part of the learning process. Working through dimension mismatches and placeholders reinforced that machine learning is not just about theory; it requires patience and attention to detail.

Overall, this workshop strengthened my understanding of image classification and neural networks. It connected the math, the code, and the practical application into one cohesive experience. It also showed me that even a basic neural network contains many moving parts that must work together correctly. Completing this lab made the concepts feel more real and less abstract, and it gave me a stronger foundation for future projects involving computer vision and deep learning.

References

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